BIOECONOMIC IMPLICATIONS OF THE INTERACTION OF MIDDLEMEN AND DISTANCE: THE CASE OF A BUSHMEAT SUPPLY CHAIN

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Abstract. Resource supply chains in low and middle income countries (LMIC) often depend on middlemen to transport the harvested resource to a local market for sale. These middlemen may drive the extraction locations and levels of the resource and therefore their decisions should be included in models of resource extraction. I model middlemen, both monopsonistic and competitive, as well as local hunters, and show that middlemen can induce harvesting in more distant locations than hunters alone, and middlemen may continue to harvest at lower stock levels. I find that monopsonistic middlemen protect resources at closer locations in a single period which is consistent with the literature. However, I also find that this result does not necessarily hold over time because these middlemen continue to harvest at lower stock levels. I apply these results to a case study of the fruit bat supply chain in Sulawesi, Indonesia to explain harvesting expansion and continued harvesting despite low population levels.

1. INTRODUCTION

In low and middle income countries (LMIC), the harvesting of resources may not occur near where the resources can be sold at market. Market access for harvesters may be limited by distance leading to arbitrage opportunities for trade intermediaries, or middlemen, to enter the market and transport resources from the point of extraction to the markets. These middlemen purchase the harvest from local villagers or local collectors and transport the harvest to the market where they sell to the vendors. Such middlemen have been observed in numerous bushmeat markets as well as other types of resource markets (*e.g.* fisheries, produce) in LMICs around the world (e.g Ngeleza and Robinson, 2013; Tháy et al., 2019; Latinne et al., 2020). These additional actors play an important role in extraction levels and locations. I develop a spatially explicit model of this middleman behavior, and using a case study from Sulawesi, Indonesia, I show how having middlemen —with and without market power —present in a system results in different harvesting outcomes across space and time compared to a system with no middlemen.

Bushmeat supply chains are of particular interest due to both conservation concerns and emerging infectious disease risk (Albers et al., 2020). Understanding the decisions of actors in a supply chain better allows us to identify and evaluate policy levers to protect species and public health. In LMICs concerns for conservation and disease risk must be balanced with the welfare of the households that rely on the wildlife trade for income and nutritional needs (Bachmann et al., 2019). Wildlife hunting and trapping may done done legally or illegally, for sale locally or abroad for food, pets, or other purposes (Fukushima et al., 2020; McNamara et al., 2019). Here I focus on hunting for bushmeat and the supply chain which brings it to local markets. Bushmeat hunting, trade, and consumption has been linked to SARS-CoV, SARS-2-CoV, HIV, and other disease outbreaks (Andersen et al., 2020; Karesh et al., 2005). Hunters come into direct contact with live wildlife creating an opportunity for disease transmission and they may not take precautions to reduce exposure when unaware

of the potential risk (Harrison et al., 2011; Wolfe et al., 2005). Bushmeat demand is largely driven by income and nutritional needs, and has varying price and income elasticities across countries (Fa et al., 2009; East et al., 2005; Godoy et al., 2010; Wilkie and Godoy, 2001; Wilkie et al., 2005).

In Sulawesi, Indonesia concerns of both over-harvesting and disease spillover risk have created an urgent need to better understand the structure of the bushmeat supply chain so that future analysis can better predict potential policy implications (Latinne et al., 2020). The levels of harvest identified by Latinne et al. (2020) indicate that the islands fruit bat (flying fox) populations are not large enough to support the level of harvesting that is occurring. Loss of these populations results in the loss of important ecosystem services including the pollination of durian which is an economically important crop in southeast Asia (Aziz et al., 2017). The hunting, transportation, and sale of flying foxes also presents a risk of spillover of an emerging infectious disease (Huong et al., 2020). Modeling the supply chain increases the understanding of the human-bat contact that is occurring.

Sulawesi presents a unique case study as the bushmeat markets are primarily located in North Sulawesi where the population is majority Christian. In the south of the island, the majority Muslim population generally does not consume bushmeat. This spatial disparity in consumption is likely what caused the flying fox populations in North Sulawesi to decline (Latinne et al., 2020). The bushmeat markets in Sulawesi have been the focus of research for decades (Clayton et al., 1997; Lee et al., 2005; Clayton and Milner-Gulland, 2000; Sheherazade and Tsang, 2015; Latinne et al., 2020). The structure of the supply chain was described in 1997 by (Clayton et al., 1997). At the time, bats were mainly harvested in Gorontalo which borders North Sulawesi. Vendors would purchase bats directly from hunters and transport the bats to the market themselves (Clayton et al., 1997). Latinne et al. (2020) collected data in November 2018 to February 2019 from the markets in North Sulawesi, and conducted interviews with vendors, middlemen, and hunters. Their research

shows that in the decades since Clayton et al. (1997), middlemen have entered the supply chain as bat hunting has moved farther from the markets. Latinne et al. (2020) find that bats are now mostly harvested in the provinces in the southernmost part of the island, and that bat populations close to the market have declined or disappeared.

I develop a spatially explicit model of local harvesters and middlemen to show how resource extraction levels and locations change under three different cases of middleman type and market power for five villages increasingly far from the market. The cases are (1) a representative local hunter in each village, with no middlemen, so that hunters transport their own harvest to the market, (2) a competitive middleman who travels to multiple villages collecting the harvest from the local representative hunter, (3) a monopsonist middleman who travels to multiple villages collecting the harvest from the local representative hunter.

2. LITERATURE REVIEW

Markets in equilibrium with middlemen are inefficient and this may cause a negative perception of middlemen (Masters, 2008). However, they are also noted for playing a vital role in some markets and their resulting inefficiencies can be minimized through policy (Arya et al., 2015). In resource markets, these inefficiencies may actually serve to protect resources from overexploitation. Tháy et al. (2019) show that middlemen with market power can protect resources from reaching an open-access outcome. The role of middlemen in resource markets in LMICs have been well documented, especially in small-scale fisheries. Findings show middlemen may provide access to credit and markets, technological expertise, and contracts that mitigate risks (e.g. Latinne et al., 2020; Russell, 1987; Aggarwal, 2008; Crona et al., 2010). However in some cases they may exploit local harvesters, capturing more of the surplus and increasing risks to local harvesters (Aggarwal, 2008). The relationships between middlemen and local producers can drive the harvesting behavior. For example, middlemen may induce

non-selective captures or illegal poaching (Nascimento et al., 2017; Pedroza, 2013). Most of this research has been survey work identifying the affect of middlemen on the local producers outcomes and harvesting decisions. Little research has been done on the decisions of middlemen and their implications for broad conservation outcomes. While the relationships between middlemen and harvesters vary across locations, understanding how the presence of middlemen can generally affect harvest levels across space and time can provide policy relevant insights. To the best of my knowledge, no previous work on middlemen has considered the relevant spatial decisions made over time.

This research also contributes to the understanding of bushmeat market structures. This is an important area of research as the world becomes more aware of emerging infectious disease risk (Albers et al., 2020). Wildlife hunting and consumption in LMICs can be important for households both nutritionally and economically. Previous economic research on wildlife hunting as focused on the time and land trade-offs faced by households involved in hunting and agriculture activities (Johannesen and Skonhoft, 2004; Bulte and Horan, 2003). Some hunted species may be a nuisance to agriculture which creates incentives for hunting that must be addressed when developing policies to protect species (Bulte and Rondeau, 2007; Johannesen, 2007). In the case of Sulawesi, the flying foxes are considered by fruit farmers to be a pest eating the fruit (Latinne et al., 2020). Policies aimed at reducing hunting can have unintended consequences when the policies increase enforcement or the opportunity cost of hunting, which can result in higher levels of protected species being harvested (Damania et al., 2005; Robinson, 2008). The presence of middlemen in bushmeat supply chains has been well established, but few models of their behavior exist (Bowen-Jones et al., 2003; Cowlishaw et al., 2005; Kamins et al., 2011; Nielsen et al., 2014; Bachmann et al., 2019; Van Vliet et al., 2019). This paper contributes to the literature by including the role of distance in middleman behavior which results in varying conservation outcomes across space. Allowing for middlemen with varying market power and spatial decisions identifies regions

where middlemen may increase or decrease harvest levels and are therefore beneficial or detrimental to resources.

3. **Model**

Middlemen help connect local hunters to a distant market. Local hunters harvest a resource within their village. Entry to hunting is limited by the village population and rents will not fully dissipate despite the resource being unregulated and unprotected. There are five equally spaced villages located on a line, where the market is located on the end of the line closest to village one, and furthest from village five (Figure 1).

If there are no middlemen, the hunters harvest the resource then must transport it to the market. When middlemen are present, the hunter harvests the resource and can then sell that harvest to the middleman at their village or they can transport the harvest to the market themselves. The price the hunter receives from the middleman will vary across space and middleman case.

3.1. **Case 1: No Middlemen**

In each village, when there are no middlemen operating, hunters must transport their own harvest to market. The hunter in each area i can harvest

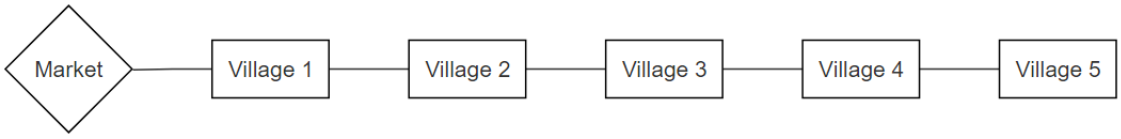

Figure 1. The spatial representation. The villages are equally spaced.

$$h_i = qE_iX_i \tag{1}$$

where E_i is the hunting effort chosen by the hunter in area i, X_i is the stock of the hunted species in area i, and q is the catchability coefficient. The hunter can sell the harvest for price P_M at the final market. The hunter's total revenue is

$$TR = P_M qE_iX_i. \tag{2}$$

The hunter faces both effort costs and transportation costs. The hunters total costs are

$$TC = \frac{\gamma}{2}E_i^2 + \omega d_i h_i + 2\eta d_i. \tag{3}$$

The hunters effort costs are increasing and convex, with γ as the marginal cost of effort parameter. Distance costs depend on the distance of village i from the market (d_i). The constant marginal cost of distance is η for the hunter. A middleman's specialization in transportation would mean him having a lower marginal cost of distance than a hunter (See appendix C). For example, a hunter would likely need to rent a vehicle to travel to the market, whereas a middleman would own a vehicle. The distance costs also increases with harvest to reflect the increased cost of a heavier vehicle, with scaling parameter ω. The cost of the return trip from the market does not depend on the size of the harvest.

There exists profits in this market because entry into the market is limited by the number of villagers. The representative hunter in each area chooses the amount of hunting effort to maximize his profits.

$$\max_{E_i} \Pi = P_M h_i - (\frac{\gamma}{2} E_i^2 + \omega d_i q E_i X_i + 2\eta d_i), \tag{4}$$

$$\text{s.t.} \quad h_i = qE_i X_i, \quad h_i \leq X_i, \quad E_i \geq 0 \tag{5}$$

with first-order condition

$$\frac{\partial \Pi}{\partial E} = P_M q X_i - \gamma E_i - \omega d_i q X_i = 0. \tag{6}$$

The representative hunter in each area chooses effort such that

$$E_i^*(X_i, d_i) = \frac{(P_M - \omega d_i) q X_i}{\gamma}. \tag{7}$$

Optimal effort decreases with distance. If the optimal level of effort would result in negative profit, then the hunter will instead chose $E_i^* = 0$ as this will result in zero profit.

The harvest under the hunter's optimal amount of effort is

$$h_i^*(X_i, d_i) = q^2 X_i^2 \left(\frac{P_M - \omega d_i}{\gamma} \right), \tag{8}$$

such that harvest level decreases with distance from the market.[1]

The representative hunter with no middlemen will be the base case used to compare how middlemen affect harvesting.

[1]The corner solution, where harvest equals stock, will occur when $P_M \geq \frac{\gamma}{q^2 X_i} + \omega d_i$. And $h_i^* = X_i$.

3.2. **Representative Hunter with Middlemen**

When middlemen are operating in the market, the hunter now has the choice to transport their own harvest or sell to the middleman. If the hunter transports their own harvest, their behavior is reflected in the Case 1 model. However if the hunter sells their harvest to the middleman, the hunter model changes to reflect a new price and no distance costs. The hunter now receives price p_i from the middleman. The hunter in village i chooses effort level E_i to maximize profit.

$$\max_{E_i} \Pi = p_i h_i - \frac{\gamma}{2} E_i^2, \tag{9}$$

$$\text{s.t.} \quad h_i = qE_i X_i, \quad h_i \leq X_i, \quad E_i \geq 0 \tag{10}$$

with first-order condition

$$\frac{\partial \Pi}{\partial E_i} = p_i q X_i - \gamma E_i = 0. \tag{11}$$

The hunter chooses effort E_i equal to

$$E_i^*(p_i, X_i) = \frac{p_i q X_i}{\gamma}. \tag{12}$$

The optimal level of effort yields the optimal harvest level in village i:[2]

$$h_i^*(p_i, X_i) = \frac{q^2 X_i^2 p_i}{\gamma}. \tag{13}$$

[2]The corner solution, where harvest equals stock, will occur when $p_i \geq \frac{\gamma}{q^2 X_i}$. And $h_i^* = X_i$.

The harvest will no longer directly depend on distance, but instead the price the middleman pays at that village.

3.2.1. *The Reservation Price.* Once a middleman is introduced to the problem, the price the hunter receives for their harvest is no longer the market price P_M and is now the price the middleman pays them, p_i. The hunter will only accept the price p_i from the middleman if they will earn an equivalent or higher profit selling to the middleman than they would transporting their own harvest. In villages closest to the market, where the hunter faces the lowest transportation costs, it is possible the hunter would be better off transporting their own harvest. The hunter will chose to transport their own harvest to market when their profit is higher in the Case 1 model than under the middleman case. I assume the hunter knows the price the middleman will pay prior to making their harvest decision, allowing them to pick the optimal harvest level dependent on the case.

This creates a minimum price the middleman must pay in a location in order to purchase the harvest in each location. If the middleman cannot pay at or above the village reservation price, he will not be able to purchase the harvest at that village. I also assume the middleman knows this reservation price when making their trip decision.

To find the price that will result in a switch from the middleman transporting the harvest to the hunter transporting the harvest, I set the hunter's optimal profit from Case 1 equal to the hunter's optimal profit under the middleman scenario, and solve for p_i.

$$P_M q E_i(X_i, d_i) X_i - \frac{\gamma}{2} E_i(X_i, d_i)^2 - \omega d_i q E_i(X_i, d_i) X_i - 2\eta d_i = p_i h_i(p_i, X_i) - \frac{\gamma}{2} E_i(p_i, X_i)^2 \quad (14)$$

$$\frac{(P_M q X_i - \omega d_i q X_i)^2}{\gamma} - \frac{\gamma}{2}\left(\frac{P_M q X_i - \omega d_i q X_i}{\gamma}\right)^2 - 2\eta d_i = p_i \left(\frac{q^2 X_i^2 p_i}{\gamma}\right) - \frac{\gamma}{2}\left(\frac{p_i q X_i}{\gamma}\right)^2 \quad (15)$$

$$p_i^R = \frac{\sqrt{(P_M - d_i\omega)^2 - 4\gamma\eta d_i}}{qX_i} \tag{16}$$

The price, p_i^R, is the minimum price the middlemen can pay at village i and still be able to collect the harvest there. Any price below p_i^R will result in the middleman collecting no harvest from village i.

3.3. **Case 2: Competitive Middleman**

The first type of middleman I consider is a competitive middleman traveling to multiple villages and earning zero profit. The middleman has constant marginal cost of distance ν, which is less than the hunter's constant marginal cost of distance η. Their costs also increase with the size of the harvest they are carrying, with the same scaling parameter ω as seen in the hunter case. Therefore the middleman drives out to the furthest village he will go to, and will collect the harvest while driving back toward the market. This middleman differs from a representative competitive middleman by the addition of parameter α which is the share of the harvest they purchase. A representative competitive middleman would not reflect the distance costs faced by each competitive middleman that visits a village because the representative competitive middleman is able to purchase the entire harvest when only paying the fixed distance cost once. In reality, each competitive middleman would need to pay the fixed distance cost to only purchase some share of the total harvest in the village. This means that costs are lower for the representative middleman which could result in the middleman inducing a higher harvest level than if fixed distance costs are accounted for for each middleman. See appendix A for detail on how the results from the representative competitive middleman case differ. Assuming the number of competitive middlemen is an imperfect solution to this problem, but I argue that it is an improvement on the representative competitive middleman as it allows us to see how every middleman

facing fixed distance costs affects the harvest outcomes. Future work should address this problem by modeling the competition of middlemen across time and space and identifying an equilibrium. In order to begin to account for the individual middlemen and their higher costs, I introduce a number ($n = 1/\alpha$) of identical competitive middlemen who purchase a constant share α of the harvest from the hunter in each village for a price (p_i) they takes as given and results in zero profit. The representative competitive middleman's case would be $\alpha = 1$.

The competitive middleman will travel to villages in which there exists a positive price (p_i) resulting in zero profit for the middleman. The middleman purchases a share α of total harvest in area i. They then sell that share of the harvest at the market for price P_M.

The total cost of the middleman's trip, where the furthest village he travels to is village n, and $i = \{1, 2, ...n\}$ is

$$TC = 2\nu d_n + \sum_{i=1}^{n} (\omega d_i + \alpha p_i) h_i(p_i, X_i), \tag{17}$$

and total revenue is,

$$TR = P_M \alpha \sum_{i=1}^{n} h_i(p_i, X_i). \tag{18}$$

The middleman chooses the distance to travel away from the market by evaluating the additional profit they can earn by traveling to the next village (*i.e.* traveling from village $i - 1$ to village i). For the competitive middleman who earns zero profit, this means they will travel to the next village if a positive price exists that will result in zero profit for that village. The cost of going to village i from village $i - 1$ is

$$C_i = 2\nu(d_i - d_{i-1}) + p_i \alpha h_i(p_i, X_i) + \omega \alpha h_i(p_i, X_i) d_i, \tag{19}$$

where $d_0 = 0$,

and the additional revenue from traveling to village i is,

$$R_i = P_M \alpha h_i(p_i, X_i). \tag{20}$$

The competitive middleman earns zero profit at each village, such that

$$\Pi_i = (P_M - p_i - \omega d_i)\alpha h_i(p_i, X_i) - 2\nu(d_i - d_{i-1}) = 0. \tag{21}$$

Substituting in for $h_i(p_i, X_i)$:

$$\left(\frac{\alpha q^2 X_i^2}{\gamma}\right) p_i^2 + \left(\frac{\omega \alpha d_i q^2 X_i^2 - P_M \alpha q^2 X_i^2}{\gamma}\right) p_i + 2\nu(d_i - d_{i-1}) = 0. \tag{22}$$

Solving this yields the price paid to hunters at each village (p_i) by the competitive middleman.

$$p_i = \frac{P_M - \omega d_i}{2} + \frac{\sqrt{\alpha(-8\nu\gamma(d_i - d_{i-1}) + \alpha q^2 X_i^2 (P_M - d_i \omega)^2)}}{2\alpha q X_i} \tag{23}$$

For the middleman to collect the harvest at the village, $p_i \geq p_i^R$. If competitive middleman's village price is less than the reservation price, then the middleman evaluates the trips total profit.

3.3.1. *Special Cases.* For the competitive middleman there are two scenarios under which he may be unable to cover the fixed and variable costs of visiting a closer village, but

can cover the costs of more distant villages. These are if the reservation price is greater than the price resulting from equation (23) and second is if stock is low in a village and therefore harvest is low. If the middleman cannot go to a village based on equation (21) (I will call these unprofitable villages), then I determine (1) if the middleman would travel to subsequent villages based on equation (21). If yes, then I (2) divide the fixed distance cost of traveling to the unprofitable village(s) equally by the number of remaining villages visited and adding it to the cost of traveling to the following village(s). I then determine if with the additional cost they will still travel to those villages. If the middleman goes to villages after the unprofitable village(s), then I (3) determine if the price at the unprofitable village can meet the reservation price when there are no fixed distance costs, or can induce enough harvest under a low harvest scenario.

$$\Pi_i = (P_M - p_i - \omega d_i)\alpha h_i(p_i, X_i) - 2\nu \frac{\text{Distance to last skipped village}}{\text{Number of villages collecting at}} = 0 \tag{24}$$

Equation (24) shows that this process increases the the later villages distance costs which results in a lower price being paid at those villages.

The price at the unprofitable village(s) is now determined by:

$$\left(\frac{\alpha q^2 X_i^2}{\gamma}\right) p_i^2 + \left(\frac{\omega \alpha d_i q^2 X_i^2 - P_M \alpha q^2 X_i^2}{\gamma}\right) p_i = 0. \tag{25}$$

$$p_i = P_M - d_i \omega \tag{26}$$

This p_i tells us the price the middleman could pay at the unprofitable village(s), if he goes to the later villages. If this new p_i is greater than the reservation price then the middleman will now collect at the unprofitable village.

Assuming the middleman divides the fixed distance costs equally across villages may distort actual middleman behavior. However this assumption is used until future work can model in more detail the entry and exit and interaction of middlemen over space and time.

3.4. **Case 3: Monopsonist Middleman**

I now consider a monopsonist middleman purchasing the total harvest in each village ($\alpha = 1$). The monopsonist middleman differs from the competitive middleman in that he can chose the price he pays for the harvest. In each village the middleman chooses the the price to pay to maximize his total profits subject to the hunter's reaction function. His maximization problem is separable by village and he picks price at each village to maximize the profit at that village.

$$\max_{p_i} \Pi = (P_M - p_i - \omega d_i) h_i(p_i, X_i) - 2\nu(d_i - d_{i-1}) \tag{27}$$

$$\text{s.t.} \quad h_i(p_i, X_i) = \frac{q^2 X_i^2 p_i}{\gamma}, \quad 0 \leq p_i \leq p_i^{MAX} \tag{28}$$

The optimal price must be greater than the reservation price p_i^R and less than the maximum price p_i^{MAX} which is the price that results in zero profit for the monopsonist. This price can be found from equation (23) where $\alpha = 1$. The maximum price would therefore also be the price paid by a representative competitive middleman. If the price cannot meet these conditions at a village, then the middleman will not travel to that village. The middleman indirectly picks the harvest level, knowing the level of harvest resulting from the chosen price.

The first-order condition for each village is

$$\frac{\partial \Pi}{\partial p_i} = \frac{q^2 X_i^2 (P_M - 2p_i - \omega d_i)}{\gamma} = 0. \tag{29}$$

Solving for p_i,

$$p_i^*(d_i) = \frac{P_M - \omega d_i}{2}. \tag{30}$$

The monopsonist's optimal price is a function of distance.[3] If $p_i^* < p_i^R$ then the monopsonist middleman will pay p_i^R as long as $p_i^R < p_i^{MAX}$.[4] At the optimal price (equation (30)) the optimal harvest level is

$$h_i^*(X_i, d_i) = q^2 X_i^2 \left(\frac{P_M - \omega d_i}{2\gamma} \right). \tag{31}$$

Comparing equation (31) to equation (8), shows that the harvest under the middleman's optimal price is half of the harvest under the hunter only case.

The monopsonist middleman's total costs and revenues for all villages visited are

$$TC = 2\nu d_n + \sum_{i=1}^{n} (\omega d_i + p_i) h_i(X_i, d_i), \tag{32}$$

and

[3]The optimal price goes to zero when $d_i \geq \frac{P_M}{\omega}$. Meaning the monopsonist middleman will not collect at any village at a distance greater than that.

[4]The monopsonist's total profits are maximized when their profit in each village is maximized. As long as p_i^R results in positive profit, then the middleman will pay it. If $p_i^R > p_i^{MAX}$ then $p_i^* = 0$ and there will be no harvest collected by the middleman in that village.

$$TR = P_M \sum_{i=1}^{n} h_i(X_i, d_i). \tag{33}$$

3.5. **Population Growth**

To simulate how the population of the hunted species evolves over time under the different middleman cases I include a logistic growth function for the stock X which includes critical depensation. Ecological threshold $\underline{X}$, defines the population level below which the population cannot recover and will move to local extinction over time. The growth function is

$$G(X_i) = r(X_i - \underline{X}_i)\left(1 - \frac{X_i}{K_i}\right). \tag{34}$$

Where r is the intrinsic growth rate and K is the local carrying capacity.

I assume neither the middlemen nor the hunters are forward looking and no dispersion of the stock occurs over time.

4. **RESULTS**

I find numerical solutions for each case and show how outcomes across time and space differ when middlemen behavior is accounted for. Table 1 provides the parameter values used in the simulation. Parameter values come either directly from data collected in Sulawesi, Indonesia on the bat supply chain, or are calibrated to reflect price and cost outcomes observed there by Latinne et al. (2020). Initial stock values are intended to reflect a starting stock on the island, and not the current levels.

Table 1. Parameters

Parameter	Value	Description
q	0.0008	percent of stock harvested by unit of effort
X_i	1000000	stock of bats in each village
$\underline{X_i}$	150000	ecological threshold below which the population collapses
K	1500000	carrying capacity
r	0.27	intrinsic growth rate
γ	1.29	marginal cost of effort parameter
P_M	1.25	price paid to hunter per bat at the market in dollars
η	20.5	distance cost parameter for hunter
ν	0.5	distance cost parameter for middlemen
ω	0.0012	distance/weight cost parameter
α	0.1	share of the harvest purchased by the competitive middleman
d_1	250	distance to market from village 1 in km
d_2	500	distance to market from village 2 in km
d_3	750	distance to market from village 3 in km
d_4	1000	distance to market from village 4 in km
d_5	1250	distance to market from village 5 in km

4.1. **Single Period Harvest Outcomes**

Middlemen influence harvest levels through the price they pay to the hunters. The monopsonist middleman pays lower prices than the competitive middleman which results in a lower harvest under the monopsonist middleman case in villages where both the monopsonist and competitive middleman would travel (Figures 3 and 2). The difference in the prices offered by the two types of middlemen is drastically reduced by the inclusion of the hunter's option to transport their own harvest. The monopsonist pays the reservation price at the first two villages instead of their own lower optimal price. This also mitigates the affect of the monopsonist as a conservation force in the first two villages. If the reservation price was not accounted for, harvest would be considerably lower under the monopsonist's optimal price. Even with the competitive middleman and monopsonist paying above the reservation price the harvest in the first two villages is lower than under the case with no middleman. In these

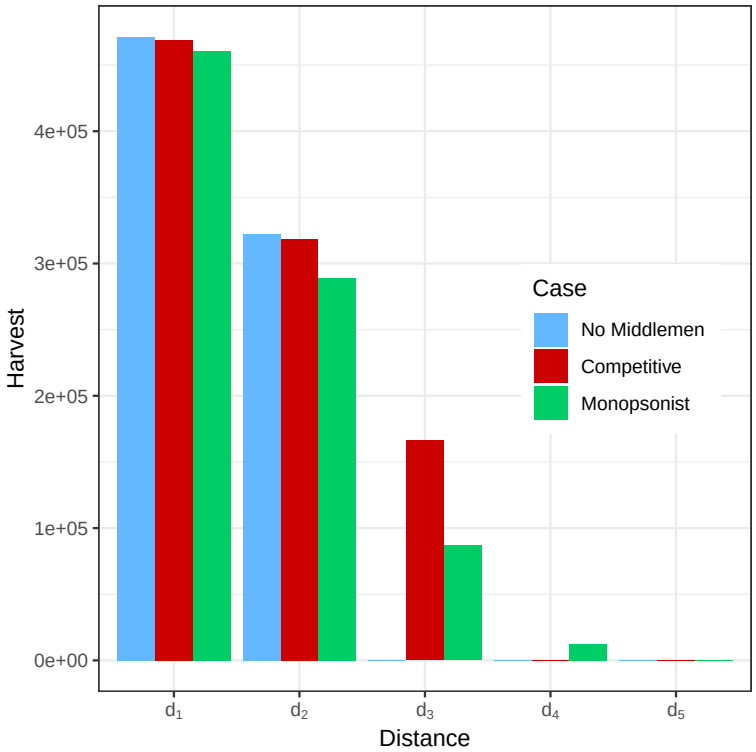

Figure 2. The harvest in each village under each case.

villages the presence of middlemen protects resources, with the monopsonist middleman offering the most protection. This is consistent with Tháy et al. (2019).

In the third village the presence of middlemen in the market is no longer good for the resource. Under the no middleman case, there would be no harvesting so far from the market. However with middlemen, there is harvesting, though monopsonist middlemen offer some protection compared to competitive middlemen. In the fourth village the monopsonist case has the worst outcome. At that distance neither competitive middlemen nor hunters without middlemen will be able to harvest profitably but the monopsonist is able to expand harvest that far. Middlemen are good for resources at the intensive margin, but cause changes of the extensive margin. Figure (4) shows the population in each village following the harvest level shown in Figure (2).

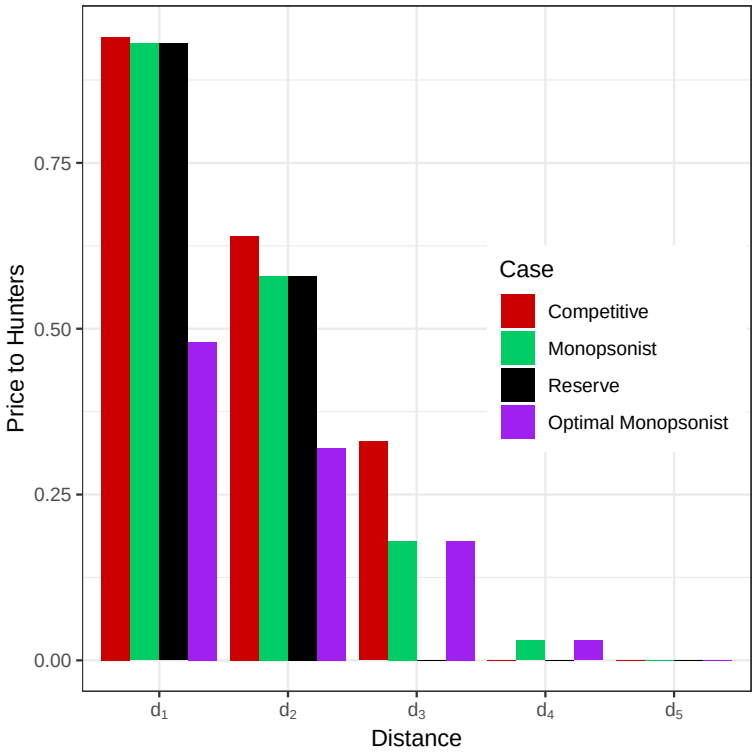

Figure 3. The price paid to hunters in each village for each case.

To compare the total impact of each case on the population, I sum the populations across villages after the harvest and show that the total population is lowest under the competitive case and highest in the no middlemen case (Figure 5). The expansion of harvesting to more distant villages caused by the presence of middlemen more than offsets the population saved in the first two villages and the middlemen harvest more total across the island than the case without middlemen.

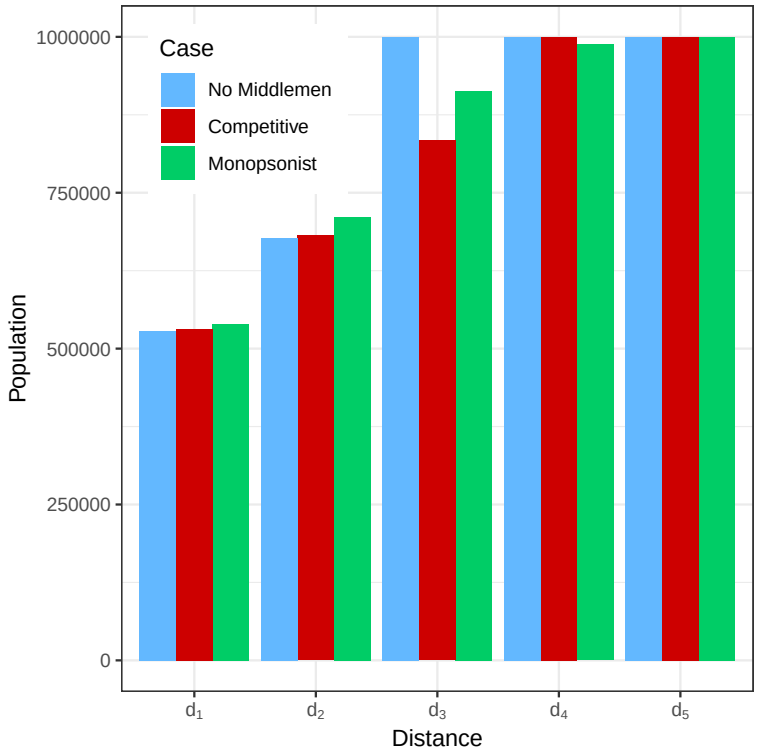

Figure 4. The population in each village for each case following the harvest.

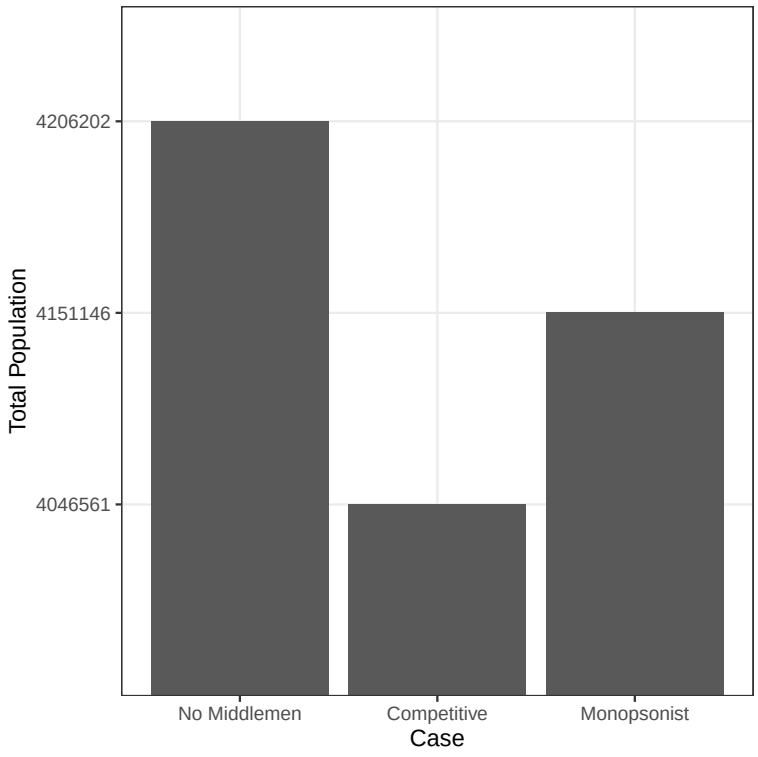

Figure 5. The total population across villages by case.

4.2. **Hunter Profits**

The impact of middlemen on local hunter profits is also an important consideration when evaluating their role in the market, especially in a LMIC setting. The hunter's profits under either middleman case will be greater than or equal to the no middlemen case in each area. The reservation price setup allows the hunter to pick their best option, and they will never allow the middlemen to make them worse off (Figure 6). As expected hunters in the first two villages are best off under the competitive middleman case and earn the same profit under the no middleman and monopsonist cases because the monopsonist pays the reservation price in those villages. In the third village both middleman cases are better for the hunter than the no middleman case as no harvest occurs in that village without middlemen. In the fourth village hunters only harvest and earn a profit under the monopsonist case.

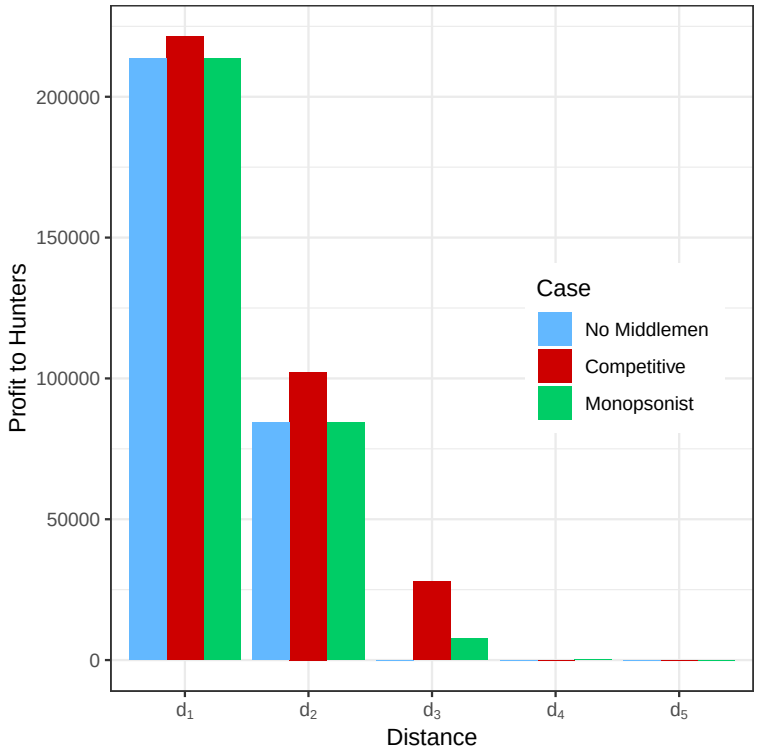

Figure 6. Hunter profit for each case in each village.

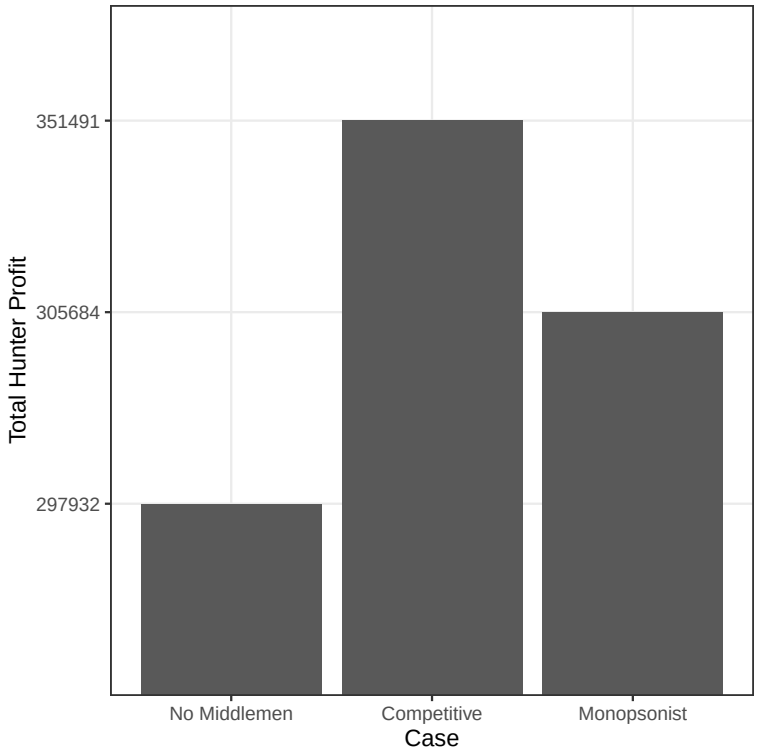

Figure 7. Total hunter profit across all areas for each case.

Summing the hunter's profits across villages shows that hunters earn the most profits under the competitive case because of the higher prices they pay, and earn more in the monopsonist case than the no middleman case because of the expanded harvesting region (Figure 7).

4.3. **Harvesting Over 50 Years**

While the model of middlemen in a single period shows their impact on harvesting levels and location, showing how these affects cause population levels and harvesting to change over time results in a better understanding in the role of middlemen in conservation outcomes. I simulate the cases over time with agents that are not forward looking and local populations that do not disperse. Figures 8 and 9 show the resulting time paths for harvesting and populations over 50 years.

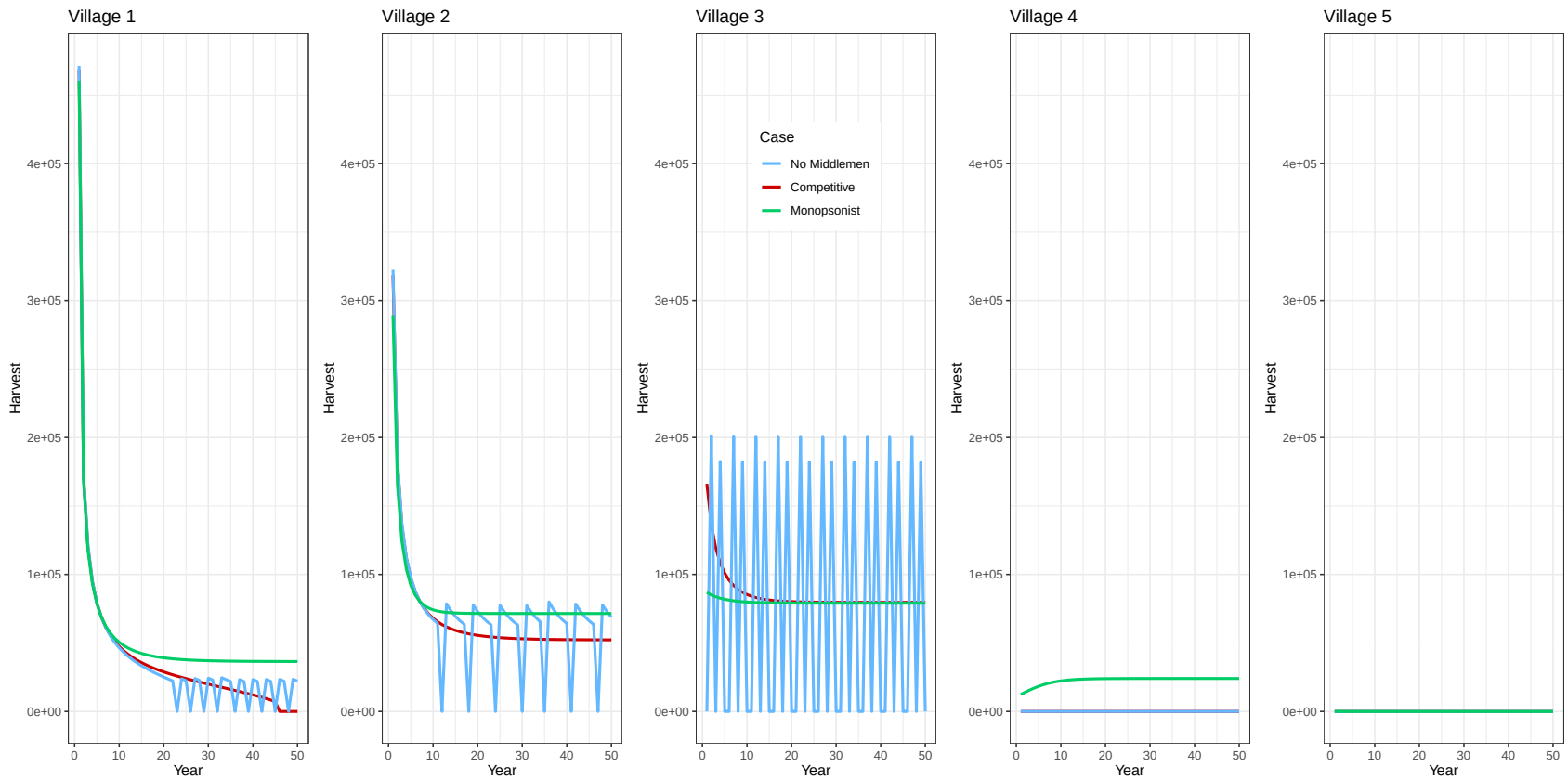

Figure 8. The harvest in each village for each case over 50 years. Neither the hunters nor the middlemen are forward looking.

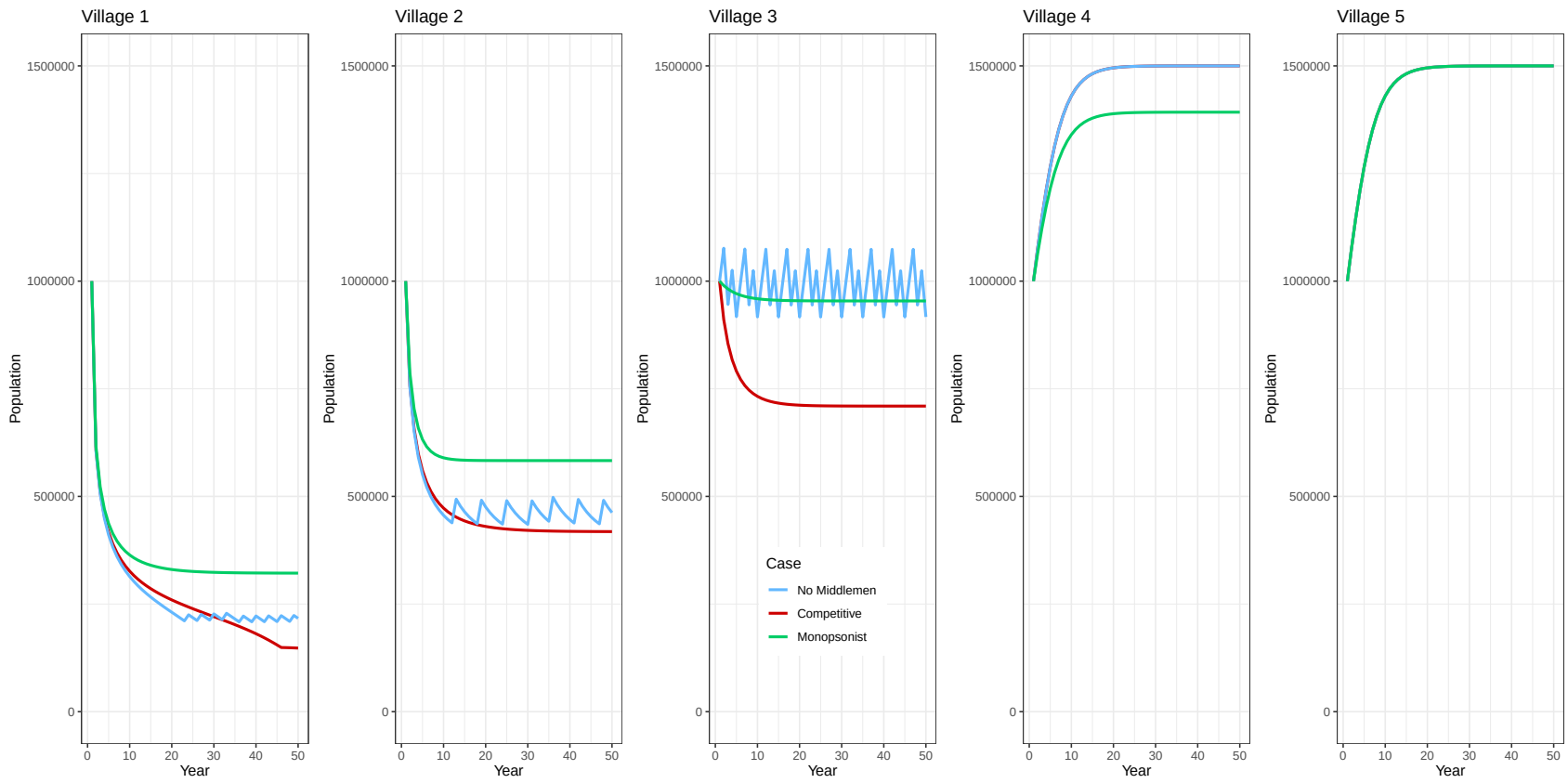

Figure 9. The population in each village for each case over 50 years. The population grows each period by the growth rate and decreased by the harvest level in each case.

Overtime, in the first two villages, the monopsonist middleman's initially lower harvest level in the first period results in a higher harvest over time at a higher population than the other two cases. Populations are also consistently lower over time for the competitive middleman case for the first three villages. This differs from the single period view of this model where competitive middlemen harvest slightly less than the no middlemen case in these villages. Competitive middlemen are able to harvest at lower stock levels than the hunters without middlemen can, this causes the competitive middleman to drive the population down to a lower level than the hunters alone will. Protections offered by competitive middlemen when comparing cases across constant stock sizes disappear when stocks fall and middlemen continue to induce harvest under conditions where hunters alone would not harvest.

4.4. **Scenario: Large Harvest**

The above case shows how lower stock sizes can change or eliminate the protective affects of middlemen. To see how the same can occur with the monopsonist middleman I raise the distance cost parameters, and increase the catchability coefficient to increase the harvest while maintaining the same initial pattern of extraction (Table 2). The first period harvest and population outcomes reflect the same role as the lower harvest level outcomes. The middlemen of both cases harvest less than the no middleman case in the closer villages, and induce new harvesting in the more distant villages (Figures 10 and 11).

Table 2. Parameters: Large Harvest

Parameter	**Value**	**Description**
q	0.0011	percent of stock harvested by unit of effort
η	85.5	distance cost parameter for hunter
ν	6.5	distance cost parameter for middlemen
ω	0.001	distance/weight cost parameter

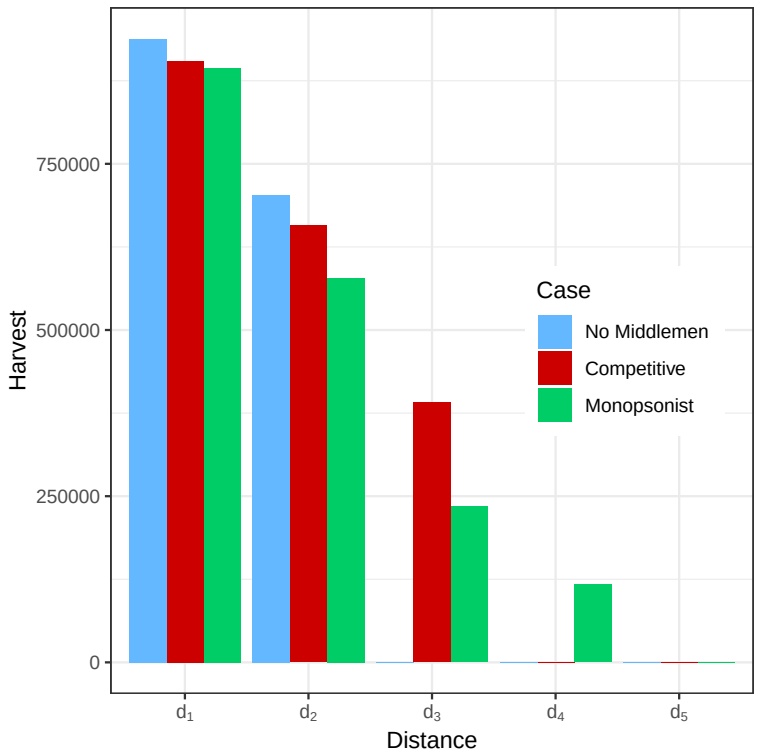

Figure 10. The harvest in each village for each case with the new parameter values.

In this scenario the local population in village one is driven to local extinction under both the monopsonist and no middleman cases. Figure 13 shows that the monopsonist is consistently worse for conservation across all outcomes. The monopsonist is able to induce harvest at the lowest population levels compared to the other cases. The monopsonist is the only case that shows consistent harvest over time (Figure 12). Under the other two cases there are years that the resource is able to recover.

The harvest becomes zero in the no middlemen case when profits from the optimal effort level become negative and optimal effort becomes zero resulting in zero harvest. The stock level at which optimal effort becomes zero is

$$\hat{X}_i^{min} = \frac{2\sqrt{d_i \eta \gamma}}{\sqrt{(P_M - d_i \omega)^2 q}} \tag{35}$$

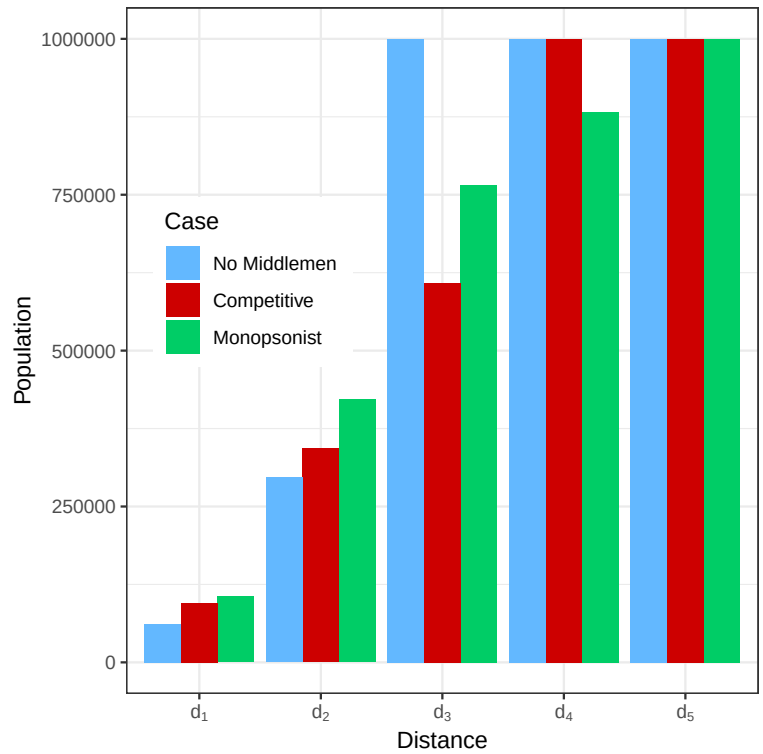

Figure 11. The resulting population in each village for each case.

The harvest becomes zero under the monopsonist case when the optimal price for the monopsonist is equal to the price that results in zero profit. The stock level at this point is

$$\tilde{X}_i^{min} = \frac{2\sqrt{2(d_i - d_{i-1})\nu\gamma}}{\sqrt{(P_M - d_i\omega)^2 q}} \tag{36}$$

The monopsonist middleman will harvest at lower stock levels if $2(d_i - d_{i-1})\nu\gamma < d_i\eta\gamma$. Recall $\nu < \eta$, and if $d_i < 2d_{i-1}$ then harvest under the monopsonist middleman case will be positive under lower stock levels. Given the parameter values for both scenarios shown, $\tilde{X}_i^{min} < \hat{X}_i^{min}$, and the monopsonist will harvest under lower stock levels in all villages. When $\tilde{X}_i^{min} < X_i \leq \hat{X}_i^{min}$, the hunter in the no middleman case will not be harvesting but the monopsonist middleman will.

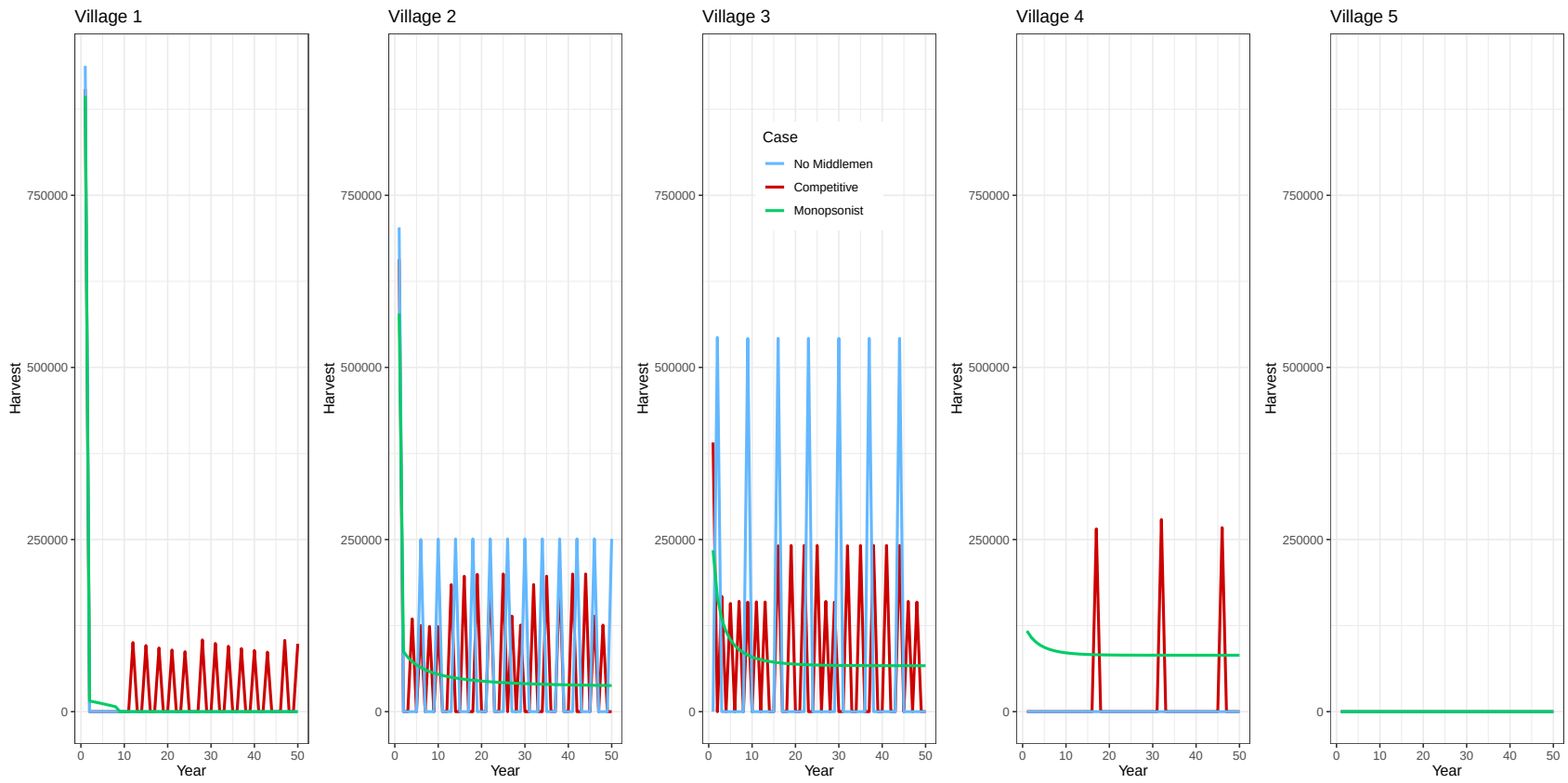

Figure 12. The harvest in each village for each case over 50 years.

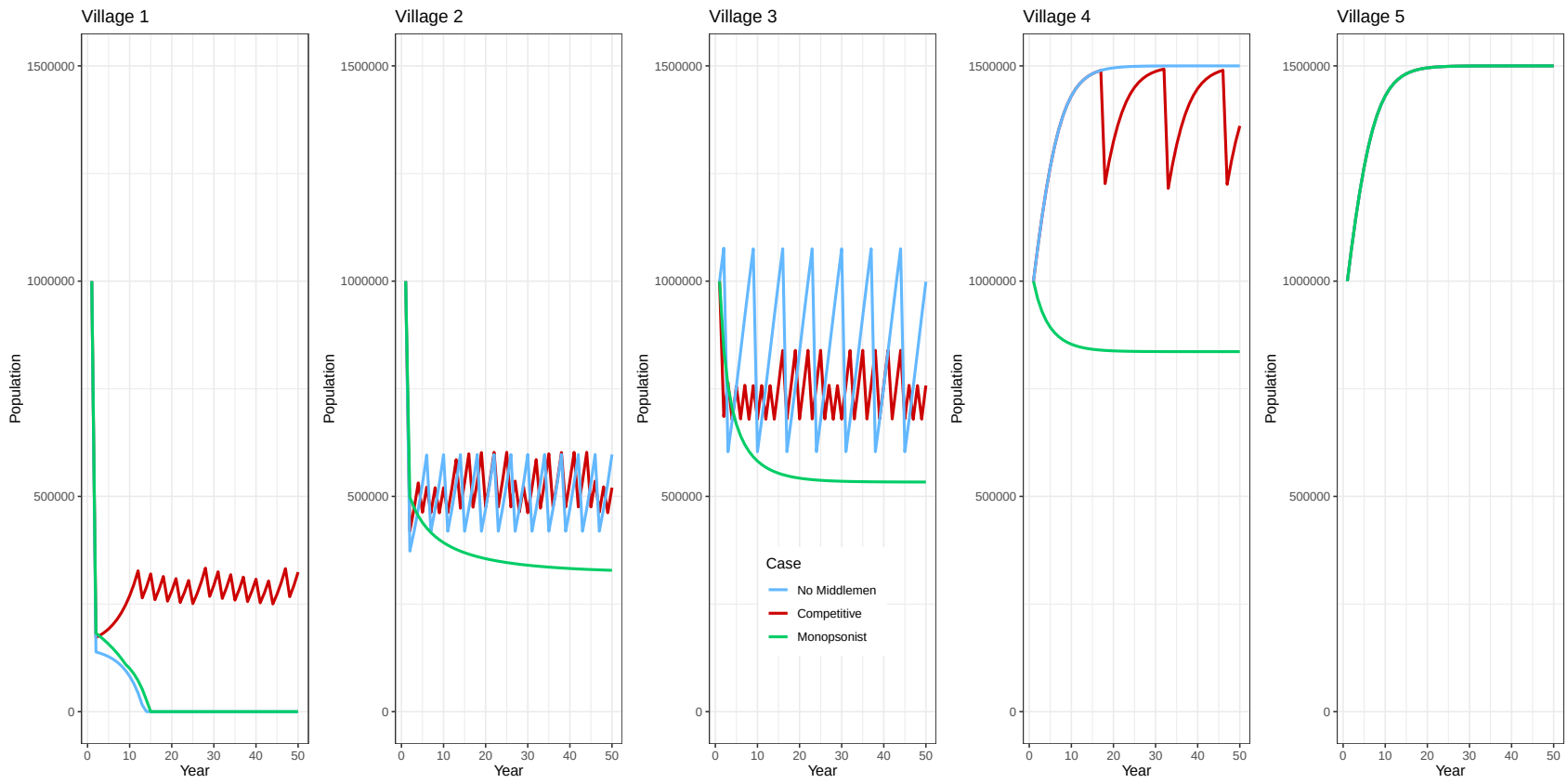

Figure 13. The population in each village for each case over 50 years. The population grows each period by the growth rate and is decreased by the harvest level in each case.

5. **Case Study: Sulawesi, Indonesia Bat Supply Chain**

This model is a good representation of Sulawesi, where middlemen travel around the island to collect bats for sale in the north. Latinne et al. (2020) estimate that 10 to 12 middlemen are operating on the island and that more than 600,000 bats are harvested annually. Understanding how these middlemen may be affecting the harvest levels observed on the island is an important step in developing policy to protect bats and public health. The model also helps explain observed changes in the market structure and bat populations over the last two decades. When no middlemen operated, harvesting was mostly done near the market and bat populations quickly declined in those areas. When populations declined close to the market, and there was a need for harvesting farther from the market, middlemen entered the supply chain and expanded harvesting to areas where hunters alone could not afford to transport the harvest to the market.

The results from this model also have implications for policy to reduce hunting of bats in Sulawesi. First it reveals the ambiguity of the impact of middlemen on resources. It is not clear that either trying to remove the middlemen entirely, or reduce or increase their market power, would benefit the areas of the island equally. Middlemen with market power are able to travel farther, even when facing high distance costs, as they can pay very little for bats. Policies aimed at increasing distance costs would work better when middlemen are more competitive. Middlemen also lead to higher profits for hunters across the island and policy to reduce the distance middlemen travel would negatively impact these hunters who would lose access to the market without the middlemen. Policy to reduce the impact of middlemen on the extensive margin could help retain the benefits they provide to populations in closer villages, while reducing their negative impacts at distant villages. Policies to provide alternative livelihoods to hunters in distant villages could help offset the loss of market access if middlemen no longer travel there.

Community resource management strategies also need to account for middlemen and the prices they can pay to induce local harvesting. Effective policy will need to address both the hunters and the middlemen. It may also be beneficial to design policy target middlemen differently at different distances from the market.

6. **Conclusion**

This model shows that middlemen play a significant role in harvesting location and levels and may lead to different population outcomes over time and space. Including the role of middlemen in resource supply chains in developing countries can explain expanding resource extraction and allow for additional policy options. Given that middlemen play such an important role, understanding their behavior is vital for understanding resource extraction in systems with middlemen. Future work on the role of middlemen should consist of considering policy options in light of this new understanding of the role of middlemen over space with a specific focus on community resource management. Fruit bats are pollinators providing benefits to fruit farmers as well as other ecosystem services. The hunting, trade, and consumption of bats also poses a risk to public health, so the benefits of reducing bat hunting are particularly salient at the moment as the world struggles with a global pandemic. While this paper is an important step in understanding the role of middlemen in developing country resource transportation, there are other relevant characteristics of these middlemen that should be considered.

Competition of middlemen varying over space could lead to more extreme price differences across location. To understand how and where competition between middlemen will develop, I plan to develop a model of individual middleman entry and exit over space and time and identify a dynamic Nash equilibrium for the number of middlemen, as well as the location strategies used. Other constraints on the middlemen will also be important in their choice

of optimal spatial strategy. Middlemen are limited in the size of the harvest they can transport in a trip by the size of their vehicle and therefore face a capacity constraint which is not accounted for in this model. Another constraint faced by the middlemen is a time constraint which limits the number of trips they can make in a year. In my current model the middleman makes one trip a year, and if travel times are negligible or time is not a binding constraint, then this model may reflect spatial decisions. However, some middlemen spend months collecting from hunters and are continuously collecting through the year. When certain locations have a longer trip time this will affect the optimal spatial strategy for the middleman.

These results are applicable to numerous supply chains in LMICs where middlemen serve a vital role of creating access to markets. The market access they create can result in worse outcomes for resources, but better outcomes for harvesters. They can also benefit resources in locations close to markets. Middlemen are an additional avenue for policy intervention and understanding how they may react to policy, especially over locations, will be key in developing effective policy to protect wildlife and prevent future pandemics.

References

Aggarwal, R. M. Resource-Poor Farmers in South India: On the Margins or Frontiers of Globalization? *Globalization and the Poor in Asia*, pages 196–220, 2008. doi: 10.1057/9780230594005_8.

Albers, H. J., Lee, K. D., Rushlow, J. R., and Zambrana-Torrelio, C. Disease Risk from Human–Environment Interactions: Environment and Development Economics for Joint Conservation-Health Policy. *Environmental and Resource Economics*, 76(4):929–944, 2020. ISSN 15731502. doi: 10.1007/s10640-020-00449-6. URL https://doi.org/10.1007/s10640-020-00449-6.

Andersen, K. G., Rambaut, A., Lipkin, W. I., Holmes, E. C., and Garry, R. F. The proximal origin of SARS-CoV-2. *Nature Medicine*, 26(4):450–452, 2020. ISSN 1546170X. doi: 10.1038/s41591-020-0820-9.

Arya, A., Löffler, C., Mittendorf, B., and Pfeiffer, T. The middleman as a panacea for supply chain coordination problems. *European Journal of Operational Research*, 240(2):393–400, 2015. ISSN 03772217. doi: 10.1016/j.ejor.2014.07.007. URL http://dx.doi.org/10.1016/j.ejor.2014.07.007.

Aziz, S. A., Clements, G. R., McConkey, K. R., Sritongchuay, T., Pathil, S., Abu Yazid, M. N. H., Campos-Arceiz, A., Forget, P.-M., and Bumrungsri, S. Pollination by the locally endangered island flying fox (Pteropus hypomelanus) enhances fruit production of the economically important durian (Durio zibethinus) . *Ecology and Evolution*, 7(21):8670–8684, 2017. ISSN 2045-7758. doi: 10.1002/ece3.3213.

Bachmann, M. E., Junker, J., Mundry, R., Nielsen, M. R., Haase, D., Cohen, H., Kouassi, J. A., and Kühl, H. S. Disentangling economic, cultural, and nutritional motives to identify entry points for regulating a wildlife commodity chain. *Biological Conservation*, 238(June):108177, 2019. ISSN 00063207. doi: 10.1016/j.biocon.2019.07.022. URL https://doi.org/10.1016/j.biocon.2019.07.022.

Bowen-Jones, E., Brown, D., and Robinson, E. J. Economic commodity or environmental crisis? An interdisciplinary approach to analysing the bushmeat trade in central and west Africa. *Area*, 35(4):390–402, 2003. ISSN 00040894. doi: 10.1111/j.0004-0894.2003.00189.x.

Bulte, E. and Rondeau, D. Compensation for wildlife damages: Habitat conversion, species preservation and local welfare. *Journal of Environmental Economics and Management*, 54(3):311–322, nov 2007. ISSN 00950696. doi: 10.1016/j.jeem.2007.02.003. URL https://linkinghub.elsevier.com/retrieve/pii/S0095069607000617.

Bulte, E. H. and Horan, R. D. Habitat conservation, wildlife extraction and agricultural expansion. *Journal of Environmental Economics and Management*, 45(1):109–127, 2003. ISSN 00950696. doi: 10.1016/S0095-0696(02)00009-8.

Clayton, L. and Milner-Gulland, E. The trade in wildlife in North Sulawesi, Indonesia. In *Hunting for sustainability in tropical forests*, pages 473–496. Columbia University Press, 2000.

Clayton, L., Keeling, M., and Milner-Gulland, E. J. Bringing home the bacon: A spatial model of wild pig hunting in Sulawesi, Indonesia. *Ecological Applications*, 7 (2):642–652, 1997. ISSN 10510761. doi: 10.1890/1051-0761(1997)007[0642:BHTBAS]2.0.CO;2. URL http://search.ebscohost.com/login.aspx?direct=true{&}db=edsagr{&}AN=edsagr.US9732796{&}site=eds-live.

Cowlishaw, G., Mendelson, S., and Rowcliffe, M. J. Structure and Operation of a Bushmeat Commodity Chain in Southwestern Ghana. *Conservation Biology*, 19(4):1194–1204, aug 2005. ISSN 08888892. doi: 10.1111/j.1523-1739.2005.00170.x. URL http://doi.wiley.com/10.1111/j.1523-1739.2005.00170.x.

Crona, B., Nyström, M., Folke, C., and Jiddawi, N. Middlemen, a critical social-ecological link in coastal communities of Kenya and Zanzibar. *Marine Policy*, 34(4):761–771, 2010. ISSN 0308597X. doi: 10.1016/j.marpol.2010.01.023.

Damania, R., Milner-Gulland, E. J., and Crookes, D. J. A bioeconomic analysis of bushmeat hunting. *Proceedings of the Royal Society B: Biological Sciences*, 272(1560):259–266, 2005. ISSN 14712970. doi: 10.1098/rspb.2004.2945.

East, T., Kümpel, N. F., Milner-Gulland, E. J., and Rowcliffe, J. M. Determinants of urban bushmeat consumption in Río Muni, Equatorial Guinea. *Biological Conservation*, 126(2): 206–215, 2005. ISSN 00063207. doi: 10.1016/j.biocon.2005.05.012.

Fa, J. E., Albrechtsen, L., Johnson, P. J., and Macdonald, D. W. Linkages between household wealth, bushmeat and other animal protein consumption are not invariant: Evidence from Rio Muni, Equatorial Guinea. *Animal Conservation*, 12(6):599–610, 2009. ISSN 13679430. doi: 10.1111/j.1469-1795.2009.00289.x.

Fukushima, C. S., Mammola, S., and Cardoso, P. Global wildlife trade permeates the Tree of Life. *Biological Conservation*, 247(March):108503, 2020. ISSN 00063207. doi: 10.1016/j.biocon.2020.108503. URL https://doi.org/10.1016/j.biocon.2020.108503.

Godoy, R., Undurraga, E. A., Wilkie, D., Reyes-García, V., Huanca, T., Leonard, W. R., McDade, T., Tanner, S., and Vadez, V. The effect of wealth and real income on wildlife consumption among native Amazonians in Bolivia: Estimates of annual trends with longitudinal household data (2002-2006). *Animal Conservation*, 13(3):265–274, 2010. ISSN 14691795. doi: 10.1111/j.1469-1795.2009.00330.x.

Harrison, M. E., Cheyne, S. M., Darma, F., Ribowo, D. A., Limin, S. H., and Struebig, M. J. Hunting of flying foxes and perception of disease risk in Indonesian Borneo. *Biological Conservation*, 144(10):2441–2449, oct 2011. ISSN 00063207. doi: 10.1016/j.biocon.2011.06.021. URL https://linkinghub.elsevier.com/retrieve/pii/S0006320711002485.

Huong, N. Q., Nga, N. T. T., van Long, N., Luu, B. D., Latinne, A., Pruvot, M., Phuong, N. T., Quang, L. T. V., van Hung, V., Lan, N. T., Hoa, N. T., Minh, P. Q., Diep, N. T., Tung, N., Ky, V. D., Roberton, S. I., Thuy, H. B., van Long, N., Gilbert, M., Wicker, L., Mazet, J. A., Johnson, C. K., Goldstein, T., Tremeau-Bravard, A., Ontiveros, V., Joly, D. O., Walzer, C., Fine, A. E., and Olson, S. H. Coronavirus testing indicates transmission risk increases along wildlife supply chains for human consumption in Viet Nam, 2013-2014. *PLoS ONE*, 15(8 August), 2020. ISSN 19326203. doi: 10.1371/journal.pone.0237129.

Johannesen, A. B. Protected areas, wildlife conservation, and local welfare. *Ecological Economics*, 62(1):126–135, 2007. ISSN 09218009. doi: 10.1016/j.ecolecon.2006.05.017.

Johannesen, A. B. and Skonhoft, A. Property rights and natural resource conservation. A bio-economic model with numerical illustrations from the Serengeti-Mara ecosystem. *Environmental and Resource Economics*, 28(4):469–488, 2004. ISSN 09246460. doi: 10.1023/B:EARE.0000036774.15204.49.

Kamins, A. O., Restif, O., Ntiamoa-Baidu, Y., Suu-Ire, R., Hayman, D. T., Cunningham, A. A., Wood, J. L., and Rowcliffe, J. M. Uncovering the fruit bat bushmeat commodity chain and the true extent of fruit bat hunting in Ghana, West Africa. *Biological Conservation*, 144(12):3000–3008, 2011. ISSN 00063207. doi: 10.1016/j.biocon.2011.09.003. URL http://dx.doi.org/10.1016/j.biocon.2011.09.003.

Karesh, W. B., Cook, R. A., Bennett, E. L., and Newcomb, J. Wildlife trade and global disease emergence. *Emerging Infectious Diseases*, 11(7):1000–1002, 2005. ISSN 10806040. doi: 10.3201/eid1107.050194.

Latinne, A., Saputro, S., Kalengkongan, J., Kowel, C. L., Gaghiwu, L., Ransaleleh, T. A., Nangoy, M. J., Wahyuni, I., Kusumaningrum, T., Safari, D., Feferholtz, Y., Li, H., Hagan, E., Miller, M., Francisco, L., Daszak, P., Olival, K. J., and Pamungkas, J. Characterizing and quantifying the wildlife trade network in Sulawesi, Indonesia. *Global Ecology and Conservation*, 21:e00887, 2020. ISSN 23519894. doi: 10.1016/j.gecco.2019.e00887. URL https://doi.org/10.1016/j.gecco.2019.e00887.

Lee, R. J., Gorog, A. J., Dwiyahreni, A., Siwu, S., Riley, J., Alexander, H., Paoli, G. D., and Ramono, W. Wildlife trade and implications for law enforcement in Indonesia: A case study from North Sulawesi. *Biological Conservation*, 123(4):477–488, 2005. ISSN 00063207. doi: 10.1016/j.biocon.2005.01.009.

Masters, A. Unpleasant middlemen. *Journal of Economic Behavior and Organization*, 68 (1):73–86, 2008. ISSN 01672681. doi: 10.1016/j.jebo.2008.03.003.

McNamara, J., Fa, J. E., and Ntiamoa-Baidu, Y. Understanding drivers of urban bushmeat demand in a Ghanaian market. *Biological Conservation*, 239(November):108291, 2019. ISSN 00063207. doi: 10.1016/j.biocon.2019.108291. URL https://doi.org/10.1016/j.biocon.2019.108291.

Nascimento, D. M., Alves, R. R. N., Barboza, R. R. D., Schmidt, A. J., Diele, K., and Mourão, J. S. Commercial relationships between intermediaries and harvesters of the mangrove crab Ucides cordatus (Linnaeus, 1763) in the Mamanguape River estuary, Brazil, and their socio-ecological implications. *Ecological Economics*, 131:44–51, 2017. ISSN 09218009. doi: 10.1016/j.ecolecon.2016.08.017. URL http://dx.doi.org/10.1016/j.ecolecon.2016.08.017.

Ngeleza, G. K. and Robinson, E. J. Cartels and rent sharing at the farmer-trader interface: Evidence from ghana's tomato sector. *Journal of Agricultural and Food Industrial Organization*, 11(1):15–30, 2013. ISSN 15420485. doi: 10.1515/jafio-2012-0011.

Nielsen, M. R., Jacobsen, J. B., and Thorsen, B. J. Factors determining the choice of hunting and trading bushmeat in the Kilombero Valley, Tanzania. *Conservation Biology*, 28(2): 382–391, 2014. ISSN 15231739. doi: 10.1111/cobi.12197.

Pedroza, C. Middlemen, informal trading and its linkages with IUU fishing activities in the port of Progreso, Mexico. *Marine Policy*, 39(1):135–143, 2013. ISSN 0308597X. doi: 10.1016/j.marpol.2012.10.011. URL http://dx.doi.org/10.1016/j.marpol.2012.10.011.

Robinson, E. J. Wanted dead and alive: To what extent are hunting and protection of an endangered species compatible? *Environment and Development Economics*, 13(5): 607–620, 2008. ISSN 1355770X. doi: 10.1017/S1355770X08004403.

Russell, S. D. Middlemen and Moneylending: Relations of Exchange in a Highland Philippine Economy. *Journal of Anthropological Research*, 43(2):139–161, jul 1987. ISSN 0091-7710. doi: 10.1086/jar.43.2.3630222. URL https://www.journals.uchicago.edu/doi/10.1086/jar.43.2.3630222.

Sheherazade and Tsang, S. M. Quantifying the bat bushmeat trade in North Sulawesi, Indonesia, with suggestions for conservation action. *Global Ecology and Conservation*, 3: 324–330, 2015. ISSN 23519894. doi: 10.1016/j.gecco.2015.01.003. URL http://dx.doi.org/10.1016/j.gecco.2015.01.003.

Tháy, P. T. T., Flaaten, O., and Skonhoft, A. Middlemen: Good for resources and fishermen? *Environment and Development Economics*, 24(5):437–456, 2019. ISSN 14694395. doi:

10.1017/S1355770X19000196.

Van Vliet, N., Muhindo, J., Nyumu, J. K., and Nasi, R. From the forest to the dish: A comprehensive study of the wildmeat value chain in Yangambi, Democratic Republic of Congo. *Frontiers in Ecology and Evolution*, 7(APR), 2019. ISSN 2296701X. doi: 10.3389/fevo.2019.00132.

Wilkie, D. S. and Godoy, R. A. Income and price elasticities of bushmeat demand in lowland Amerindian societies. *Conservation Biology*, 15(3):761–769, 2001. ISSN 08888892. doi: 10.1046/j.1523-1739.2001.015003761.x.

Wilkie, D. S., Starkey, M., Abernethy, K., Effa, E. N., Telfer, P., and Godoy, R. Role of prices and wealth in consumer demand for bushmeat in Gabon, Central Africa. *Conservation Biology*, 19(1):268–274, 2005. ISSN 08888892. doi: 10.1111/j.1523-1739.2005.00372.x.

Wolfe, N. D., Daszak, P., Kilpatrick, A. M., and Burke, D. S. Bushmeat hunting, deforestation, and prediction of zoonotic disease emergence. *Emerging Infectious Diseases*, 11(12): 1822–1827, 2005. ISSN 10806059. doi: 10.3201/eid1112.040789.

Appendices

A. **Representative Competitive Middleman**

To show how the introduction of the α parameter into the competitive middleman's model changes the outcomes for the competitive case, I show the harvest (Figure 14), population (Figure 15), and price (Figure 16) outcomes for a representative competitive middleman versus the competitive middleman modeled here (called individual for the sake of this explanation).

The harvest levels are higher for the representative competitive middleman because of the lower distance costs. The difference in harvest levels is small, but in village four the individual

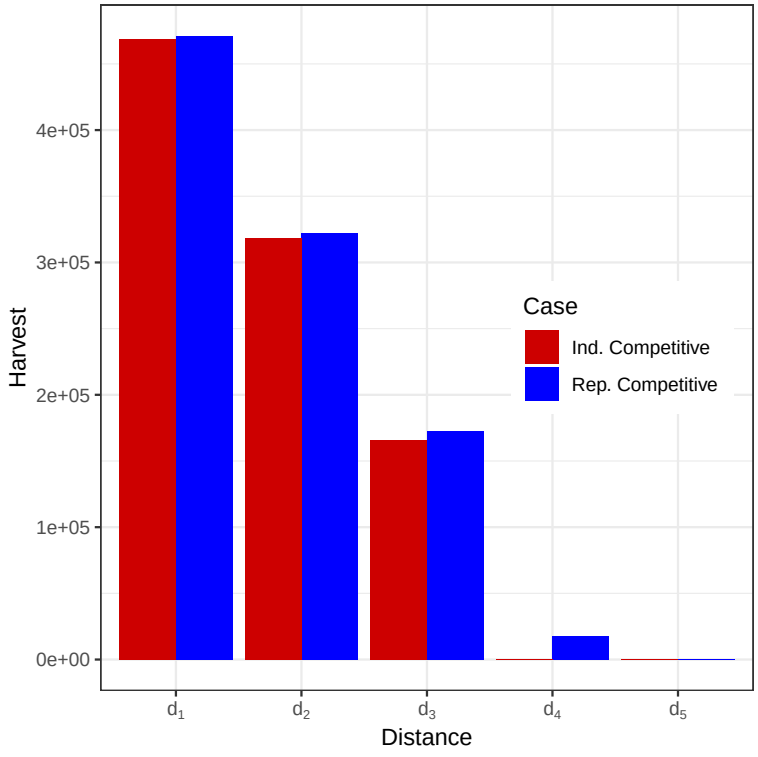

Figure 14. The harvest in each village for the representative competitive ($\alpha = 1$) versus the more individual middleman ($\alpha = 0.1$).

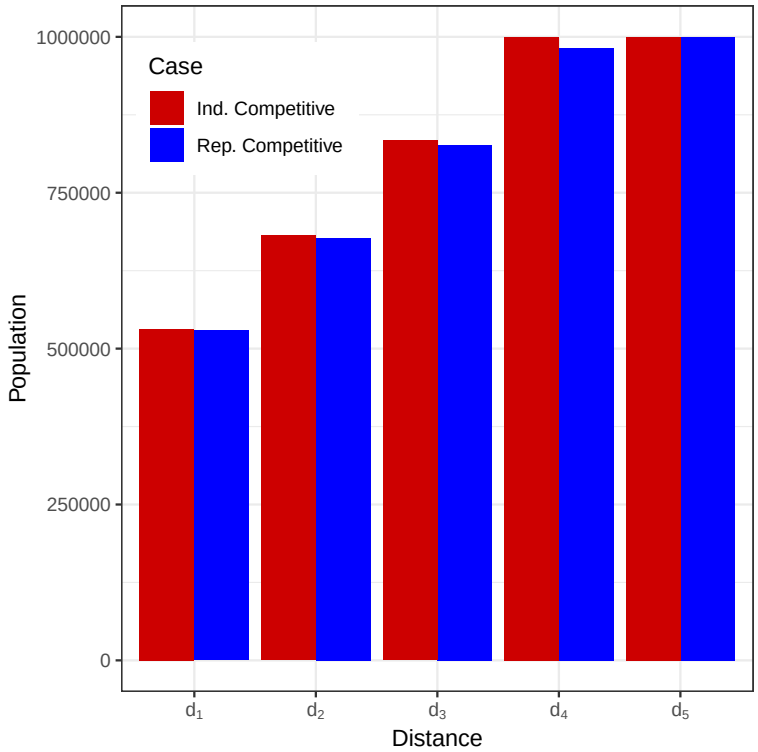

Figure 15. The population in each village for the representative competitive ($\alpha = 1$) versus the more individual middleman ($\alpha = 0.1$).

competitive middleman is not able to harvest at all while the representative competitive middleman is able to harvest there. The difference in these outcomes becomes more apparent when the middlemen face higher distance costs (See appendix C).

Without modeling the entry of middlemen over space to determine the number of competitive middlemen, which may vary across village, this individual competitive middleman model is imperfect as it assumes a fixed number of competitive middlemen for all villages. However, the representative competitive model is also imperfect and does a poor job of including distance costs. The difference between the outcomes for the two models under the original parameter set is small.

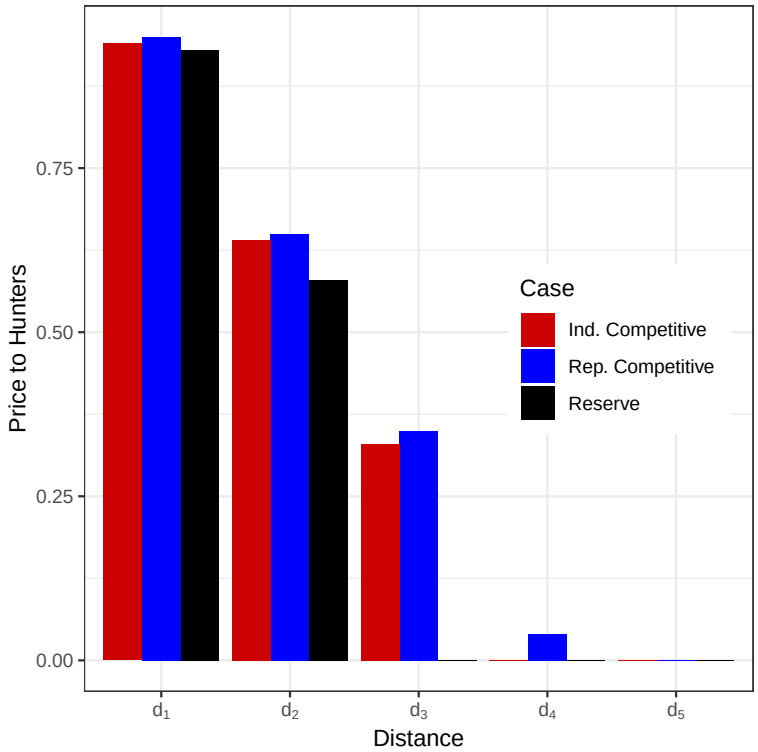

Figure 16. The price in each village for the representative competitive ($\alpha = 1$) versus the more individual middleman ($\alpha = 0.1$).

B. **Sensitivity of Number of Competitive Middleman**

The number of identical competitive middlemen assumed (n) drives the distance and harvest levels (through price) that occurs in each village. The share of the harvest that the middleman collects ($\alpha = \frac{1}{n}$) changes the maximum distance the competitive middleman travels. I vary α and find the farthest village the competitive middlemen collect at. Table 3 shows the levels of which cause a change in the last village.

Table 3. The minimum share (α) of the harvest the competitive middleman can collect and still go as far as each village. This translates to the maximum number of competitive middlemen that could be in the market and have the middleman go as far as that village.

Min. α	**Max.** n	**Farthest Village**
0.025	40	3
0.807	1	4

C. **Sensitivity of Distance Cost Parameters**

I assume that $\eta > \nu$, meaning that the hunters pay a higher per kilometer price to transport their harvest to market than the middlemen. However it is important to note the result that middlemen expand harvest distance can still be shown if this assumption does not hold. Sensitivity analysis of the farthest village where harvest occurs under each case while varying η and then ν, shows that the monopsonist middleman will travel further than the hunter, even for a range of η where $\eta < \nu$.

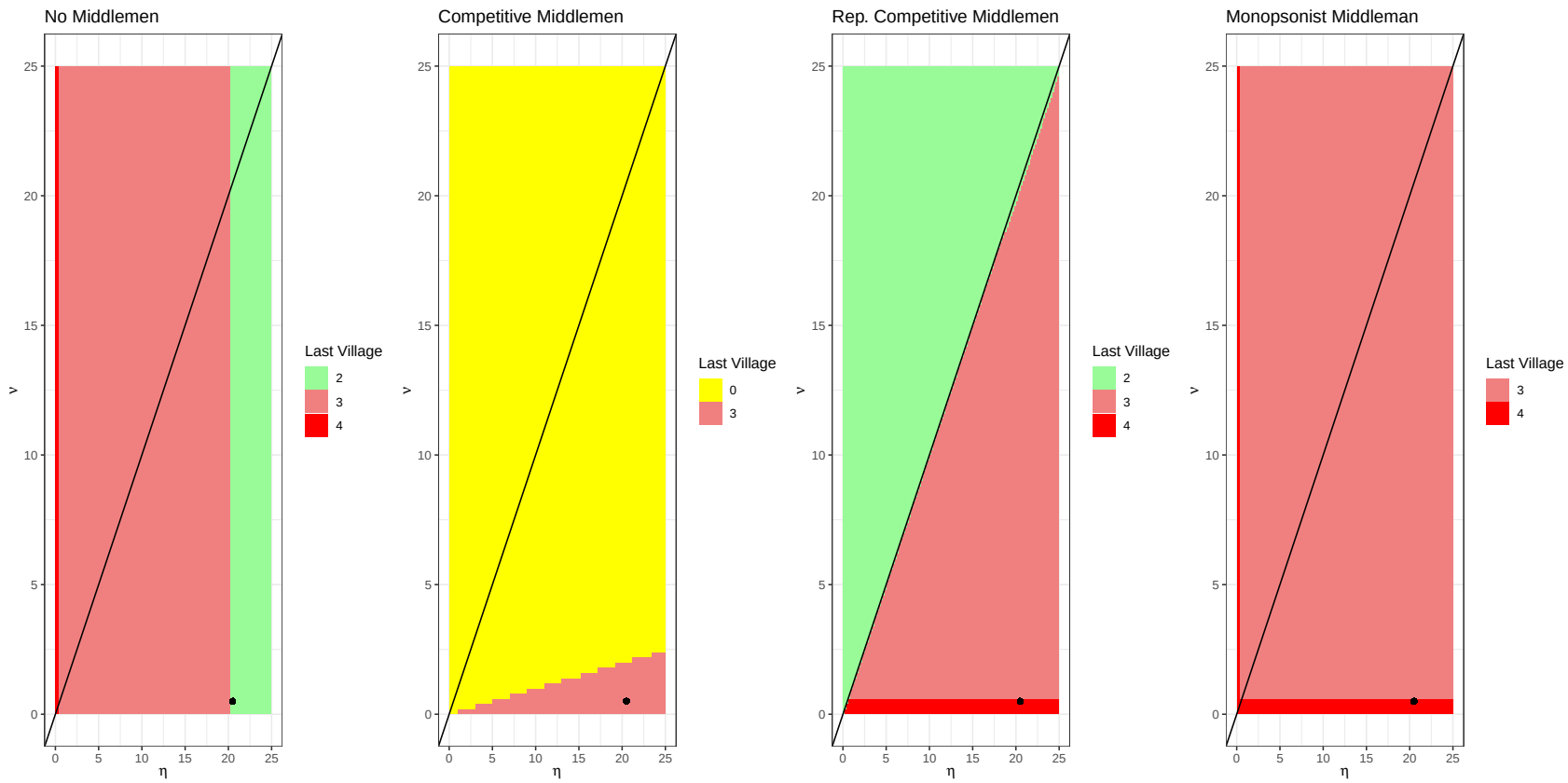

Figure 17. The farthest village harvesting occurs at under each set of cost parameters η and ν, where η is the marginal distance cost of hunters, and ν is the marginal distance cost of middlemen. The point on the graph shows the parameter combination used for the numerical solution. The line on each graph represents the points where $\eta = \nu$. Any point below this line meets the assumption $\eta > \nu$

Figure 17 also shows a significant difference between the representative and individual competitive middlemen cases. The middlemen's farthest village depends on the hunter's marginal cost of distance parameter through the reservation price. The individual competitive middleman is unable to go to any villages for a larger range of low values of η. The representative competitive middleman pays a higher price and is able to pay at or above the reservation price for lower values of η. The difference between the representative and competitive middlemen cases is most obvious when the middleman marginal cost of distance parameter, ν, is varied. Figure 17 shows that the representative competitive middleman model looks more like the monopsonist middleman.